Technology Stack

StaffSync - Modern Human Resource Management Solution

Project	StaffSync - Modern Human Resource Management Solution
Student	Neha Patidar
Enrollment No.	EN24CS3010683
Domain	Information Technology
Stack	MongoDB, Express.js, React.js, Node.js (MERN)

Technology Stack

Layer	Technology	Description
Frontend	React.js (Vite), Bootstrap 5, Axios	Responsive single page application and API calls
Backend	Node.js, Express.js	REST API, routing, middleware
Database	MongoDB, Mongoose	NoSQL storage and schema modelling
Authentication	JWT, bcryptjs	Token based login and password hashing
Dev tools	VS Code, Postman, Git, GitHub	Coding, API testing, version control
Deployment	MongoDB Atlas, Render, Vercel	Cloud database, backend hosting, frontend hosting

Application Characteristics

No.	Characteristic	Description	Technology
1	Open-source frameworks	React, Express, Node.js, Mongoose	MERN
2	Security	JWT, hashing, role-based middleware	jsonwebtoken, bcryptjs
3	Scalable architecture	Three-tier, modular routes and models	Express, MongoDB
4	Availability	Cloud hosted services	Atlas, Render, Vercel
5	Performance	Asynchronous JavaScript, indexed unique email	async/await, Mongoose